

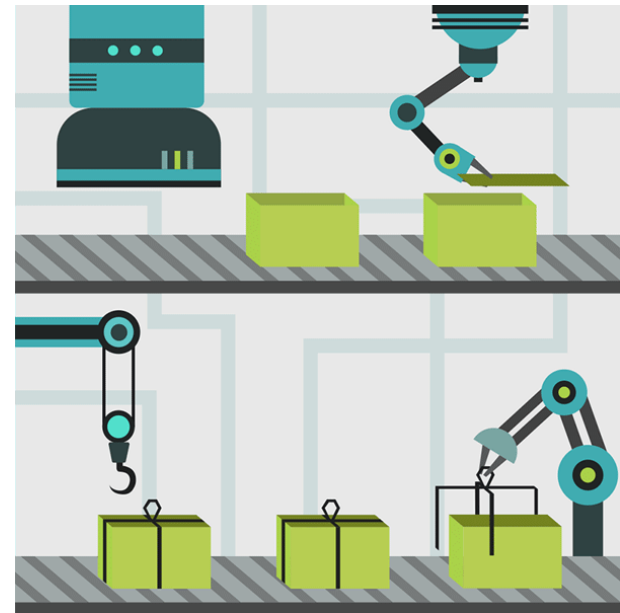
Automation



- Automation is the use of technology to perform tasks with reduced human assistance
- Automation is the technology by which a process or procedure is accomplished without human assistance



- Automation can be applied in a wide variety of areas, it is most closely associated with the manufacturing industries
- Highly automated systems, there is virtually no human participation
 - automated machine tools that process parts
 - transfer lines that perform a series of machining operations
 - automated assembly systems
 - manufacturing systems that use industrial robots to perform processing or assembly operations
 - automatic material handling and storage systems to integrate manufacturing operations
 - automatic inspection systems for quality control



ELEMENTS OF AUTOMATED SYSTEM

- Automated system implemented using a *program of instructions* combined with a *control system* that executes the instructions.
- To automate a process, *power* is required, both to drive the process itself and to operate the program and control system

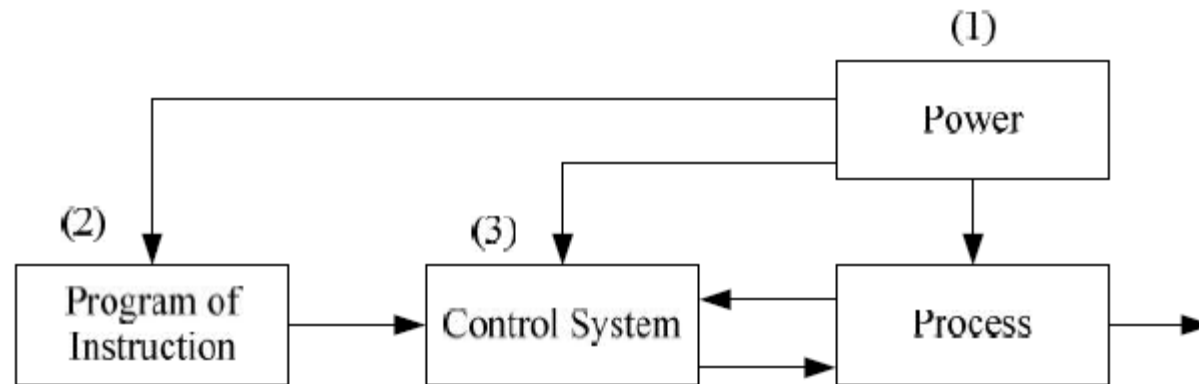


Figure 1.1 Elements of an automated system

Power

An automated system is used to operate some process, and power is required to drive the process as well as the controls. The principal source of power in automated systems is electricity.

Program of Instructions

The actions performed in an automated process are defined by a program of instructions.

POI consist of :

- **Work Cycle Programs:** *The processing steps for the work cycle* are specified in a work cycle program. work cycle consists of essentially *one step, which is to maintain a single process parameter*. However, the system becomes *complicated consisting of multiple steps with more number of process parameters* are required to control the system
- **Process parameters:** *are inputs to the process* (process variables) such as temperature setting of a furnace etc.

Control System

The control system of the automated system executes the program of instructions. *The control system causes the process to accomplish its defined function which is to carry out some manufacturing operation.*

The controls in an automated system can be either ***closed loop (feedback control system)*** and ***open loop***

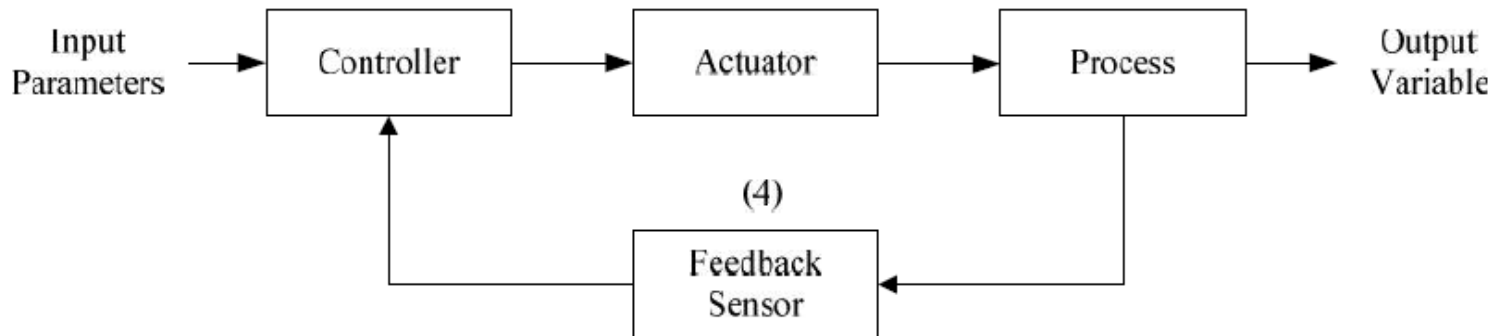


Figure 1.2 A feedback control system

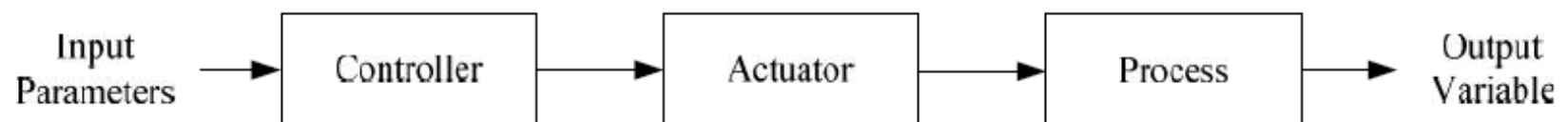


Figure 1.3 An open loop control system

Levels of automation

Levels of automation can be classified based on the complexity of the system being automated. Here's a breakdown of the different levels

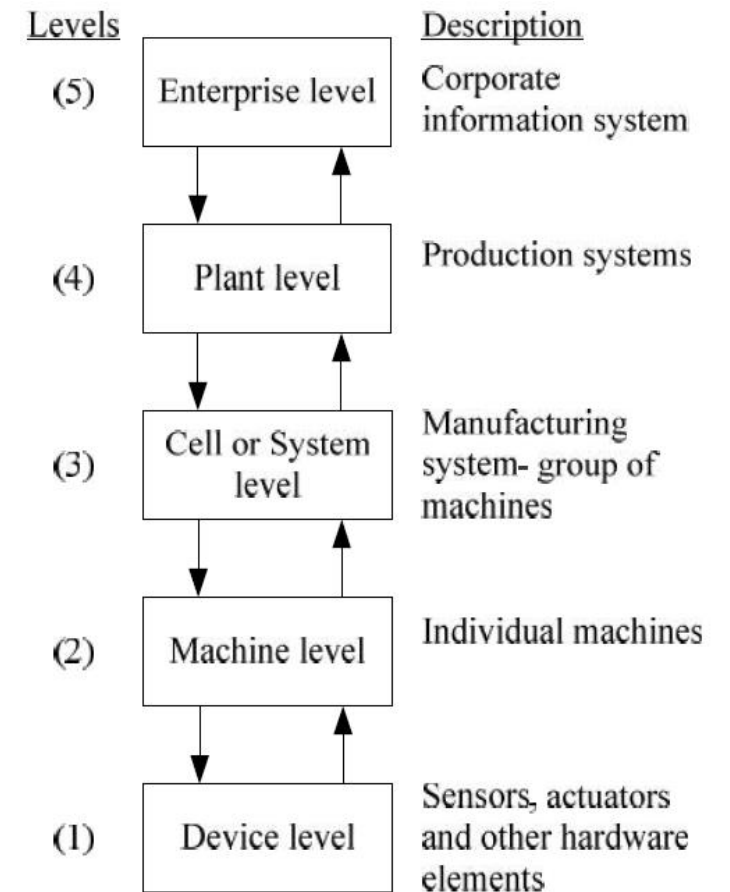


Figure 1.4 Level of automation

1. Device Level Automation

- This is the most basic level of automation, where individual devices are automated to perform specific tasks.
- Examples include automatic thermostats, light dimmers, and self-cleaning ovens.
- In this level, sensors collect data on the environment (e.g., room temperature) and send signals to controllers (e.g., thermostat) to adjust settings (e.g., adjust furnace output) to maintain a desired state (e.g., comfortable room temperature).

2. Machine Level Automation

- *This level involves automating a single machine to perform a sequence of operations.*
- Programmable logic controllers (PLCs) are typically used to control these machines.
- Examples include industrial robots in assembly lines, CNC machines, and automated packaging machines.
- Here, the automation is more complex than device level. Machines can perform a series of steps without human intervention. Sensors provide feedback on the machine's operation

3. Cell or System Level Automation

- This level automates a group of interconnected machines or devices working together to achieve a common goal
- It involves coordinating the operation of multiple machines and may involve material handling systems and automated transfer mechanisms.
- Examples include robotic welding cells, automated painting systems, and automated assembly lines.
- This level integrates multiple machines into a system that can perform a complex task. Communication and coordination between machines are crucial for efficient operation.

4. Plant Level Automation

- This level involves automating the entire production process within a factory or plant.
- It integrates all the automation systems at the cell or system level, along with production planning, scheduling, and inventory management systems.
- Plant level automation provides a holistic view and control over the entire production process. It optimizes production efficiency, resource allocation, and ensures smooth operation throughout the plant

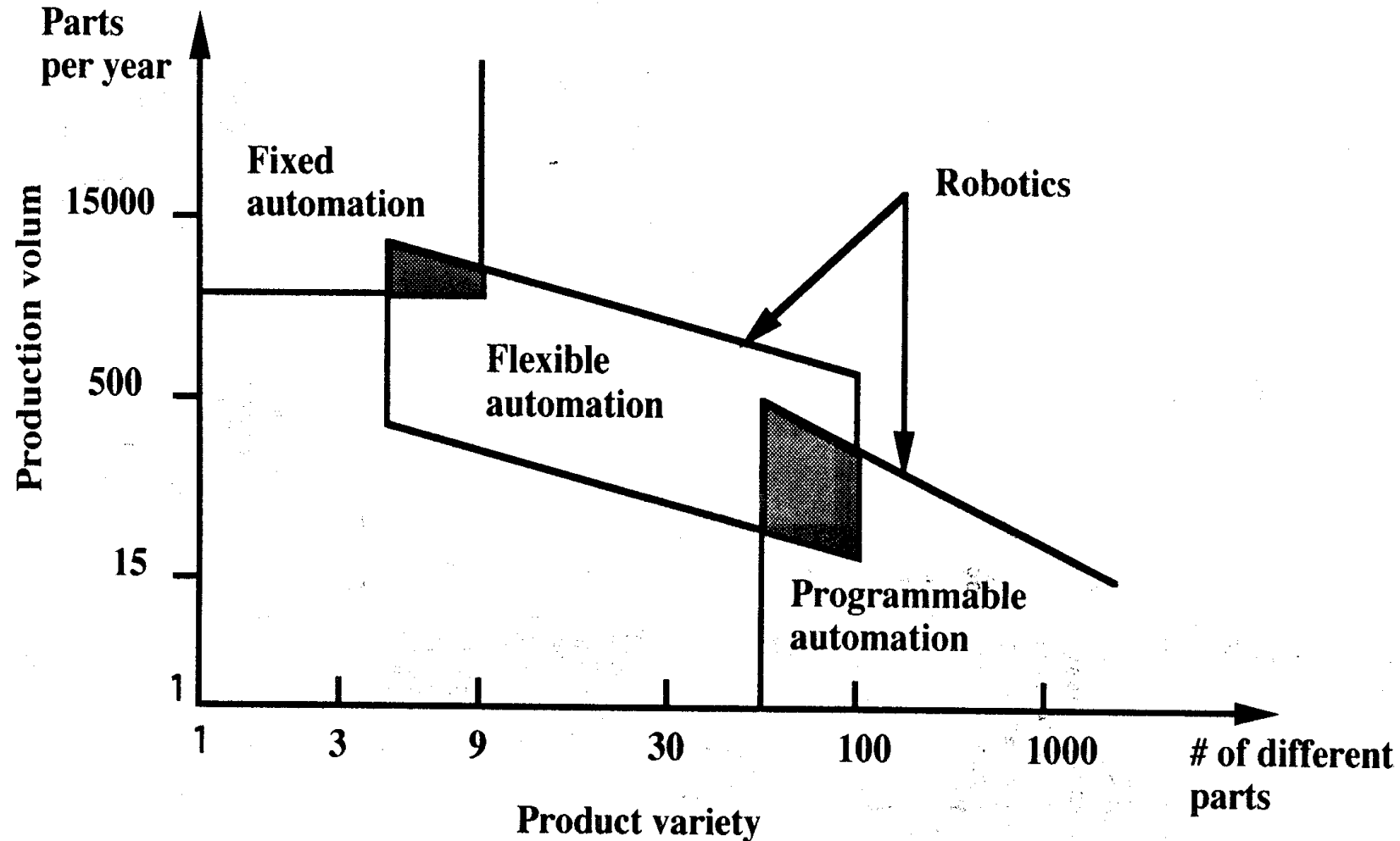
5. Enterprise Level Automation

- This is the highest level of automation, where all the automation systems across an entire enterprise are integrated.
- It includes not only the production process but also other enterprise functions such as supply chain management, customer relationship management (CRM), and enterprise resource planning (ERP).
- Enterprise level automation provides an integrated view and control over all aspects of the business.

Three Basic Types of Automation

- **Fixed automation** - the processing or assembly steps and their sequence are fixed by the equipment configuration
- **Programmable automation** - equipment is designed with the capability to change the program of instructions to allow production of different parts or products
- **Flexible automation** - an *extension of programmable automation* in which there is virtually no lost production time for setup changes or reprogramming

Three Basic Types of Automation



Features of Fixed Automation

- High initial investment for specialized equipment
- High production rates
- *The program of instructions cannot be easily changed because it is fixed by the equipment configuration*
 - Thus, little or no flexibility to accommodate product variety





Features of Programmable Automation

- High investment in general purpose equipment that can be **reprogrammed**
- *Ability to cope with product variety by reprogramming the equipment*
- Suited to batch production of different product and part styles
 - Lost production time to reprogram and change the physical setup
- *Lower production rates than fixed automation*

Features of Flexible Automation

- High investment cost for custom-engineered equipment
- *Capable of producing a mixture of different parts or products without lost production time for changeovers and reprogramming*
 - Thus, continuous production of different part or product styles
- *Medium production rates*



AUTOMATED FLOW LINES

- An automated flow line consists of several machines or workstations which are linked together by work handling devices that transfer parts between the stations
- A raw work-part enters *one end of the line* and the processing steps are performed sequentially as the part moves from one station to the next
- It is possible to incorporate buffer storage zones into the flow line, either at a single location or between every workstation.
- It is also possible to include inspection stations in the line to automatically perform intermediate checks on the quality of the work-parts

objectives of the use of flow line automation

- To reduce labor costs
- To increase production rates
- To reduce work-in-process
- To minimize distances moved between operations
- To achieve specialization of operations
- To achieve integration of operations

Configurations of automated flow line - In-line type

- The in-line configuration consists of a sequence of workstations in a more or less straight line arrangement*

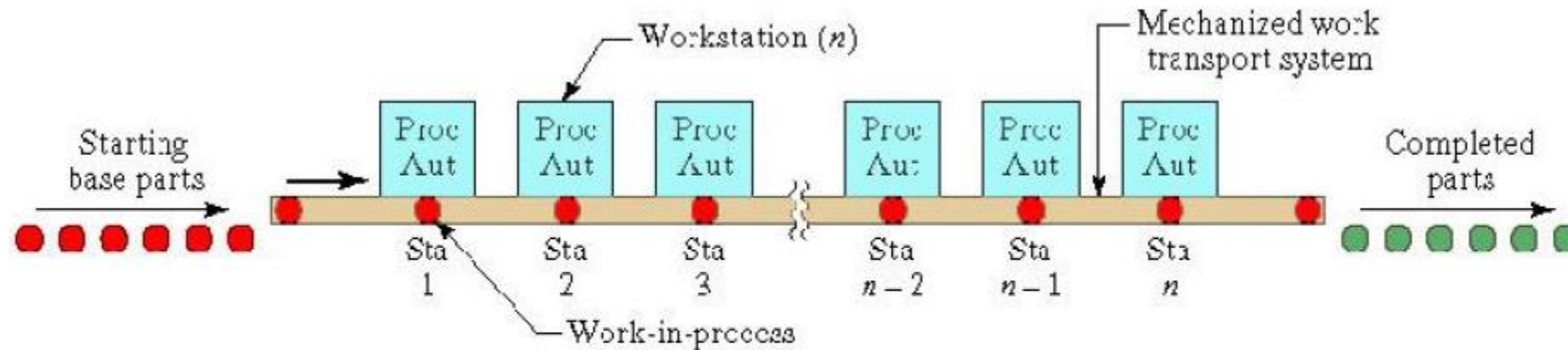


Figure 1.8 Flow line (Inline type)

Segmented In-Line Type

- The segmented in-line configuration consists of two or more straight line arrangement which are usually perpendicular to each other with L shaped or U shaped or rectangular shaped
- *The flow of work can take a few 90° turns, either for workpiece reorientation, factory layout limitations*

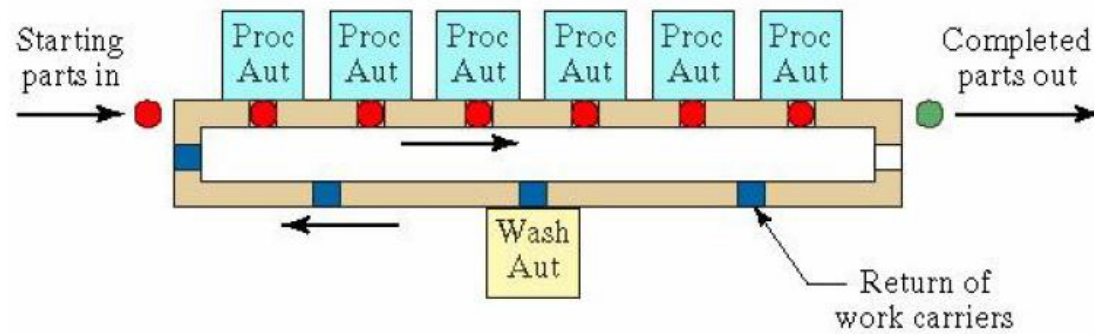


Figure 1.14 Rectangular-shaped configuration

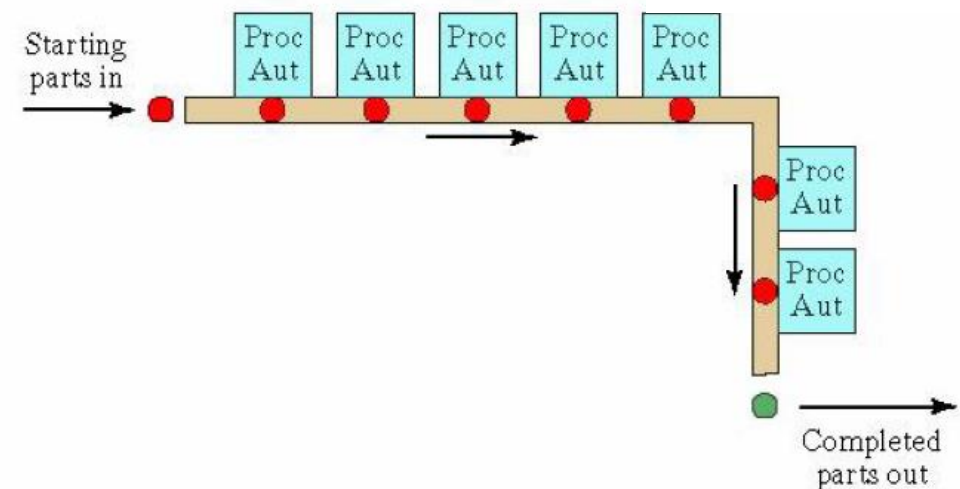


Figure 1.12 L-shaped configuration

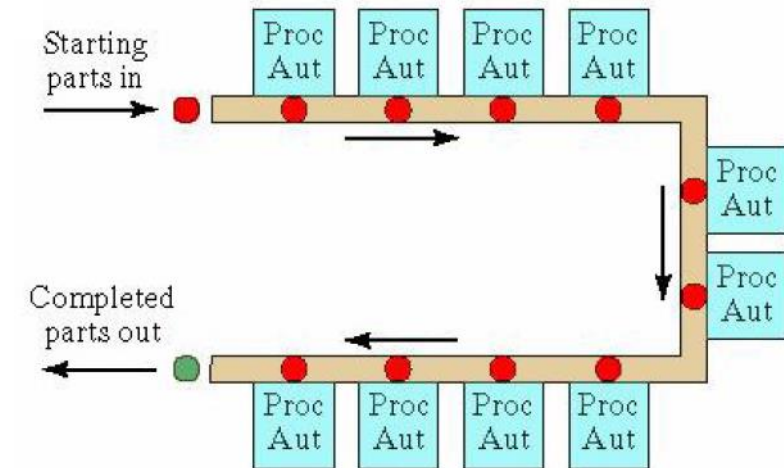


Figure 1.13 U-shaped configuration

Rotary type

- The work-parts are indexed around a circular table or dial
- The workstations are stationary and usually located around the outside periphery of the dial.
- The parts ride on the rotating table and positioned, in turn, at each station for its processing or assembly operation.

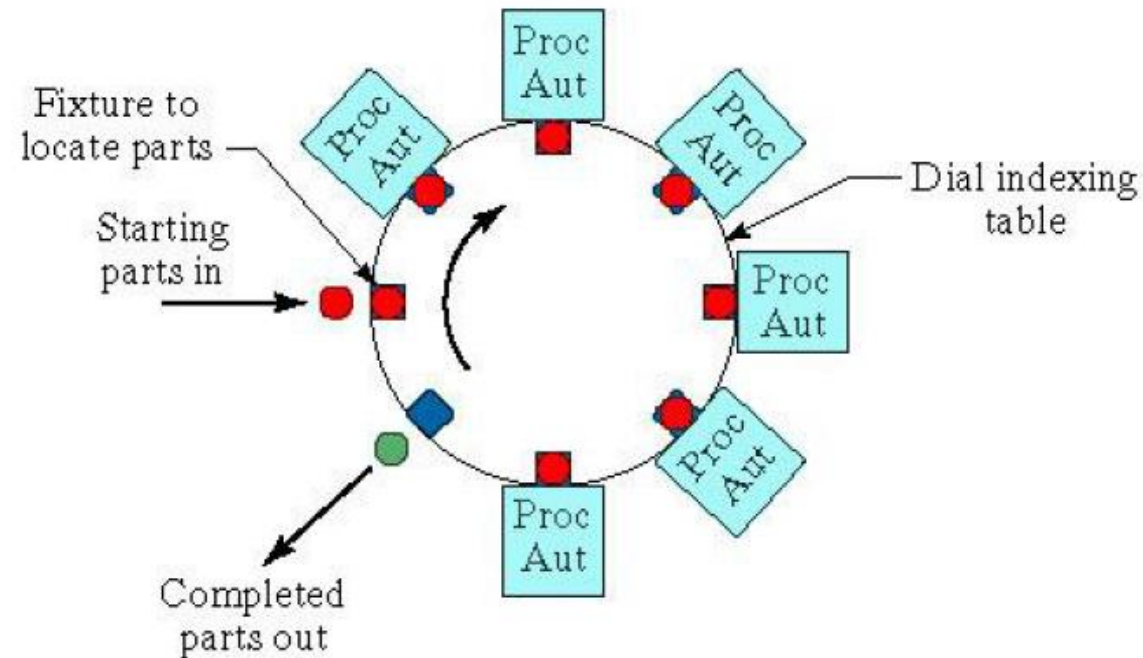


Figure 1.15 Rotary configuration

Transfer mechanisms

- To move parts between stations
- These mechanisms can be grouped into two types:

Linear travel for in-line machines

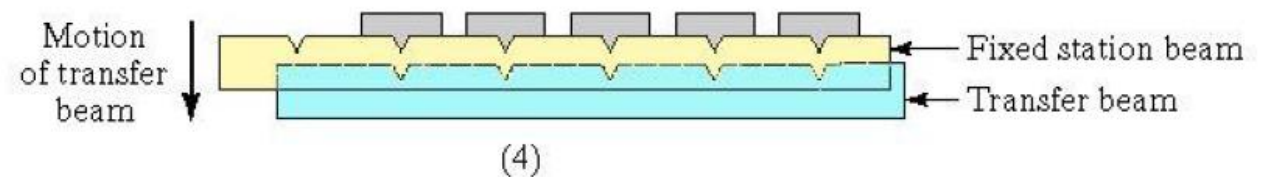
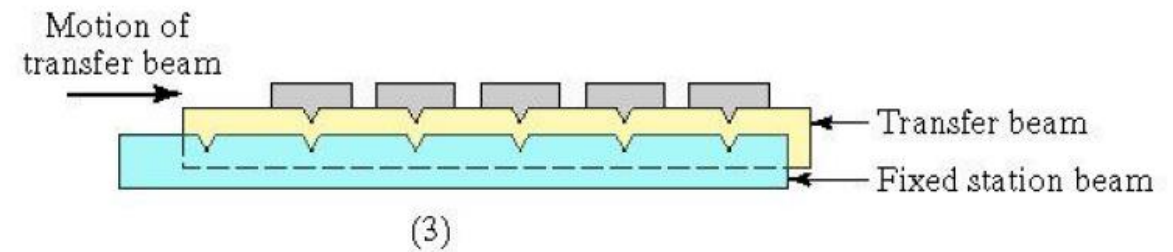
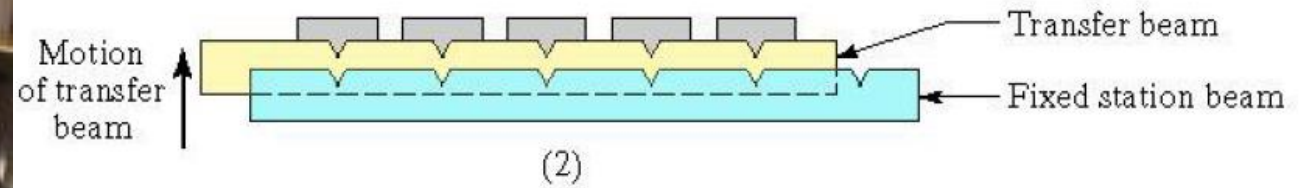
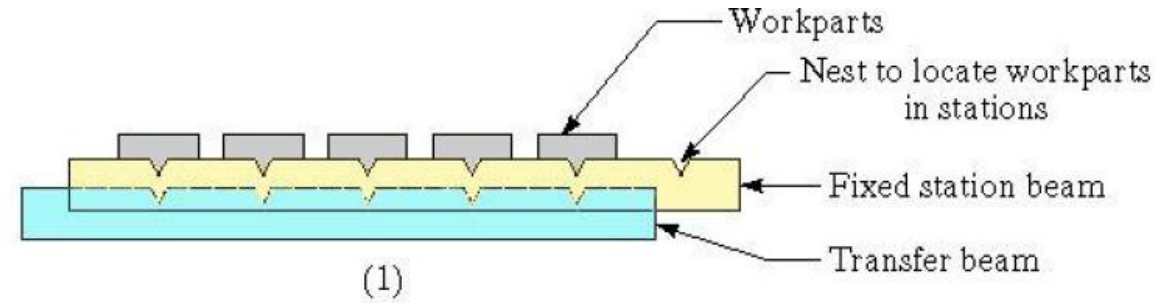
Rotary motion for dial indexing machines

Linear transfer mechanisms

a. Walking beam systems

- The work-parts are lifted up from their workstation locations by a transfer bar and moved one position ahead, to the next station.
- The transfer bar then lowers the parts into nests which position them more accurately for processing.
- The motion of the beam is generated by a rotating camshaft powered by an electric motor

a. Walking beam systems



b. Powered roller conveyor system

- The conveyor can be used to move pans or pallets possessing flat riding surfaces
- The rollers can be powered by either of two mechanisms.

Belt drive

Chain drive

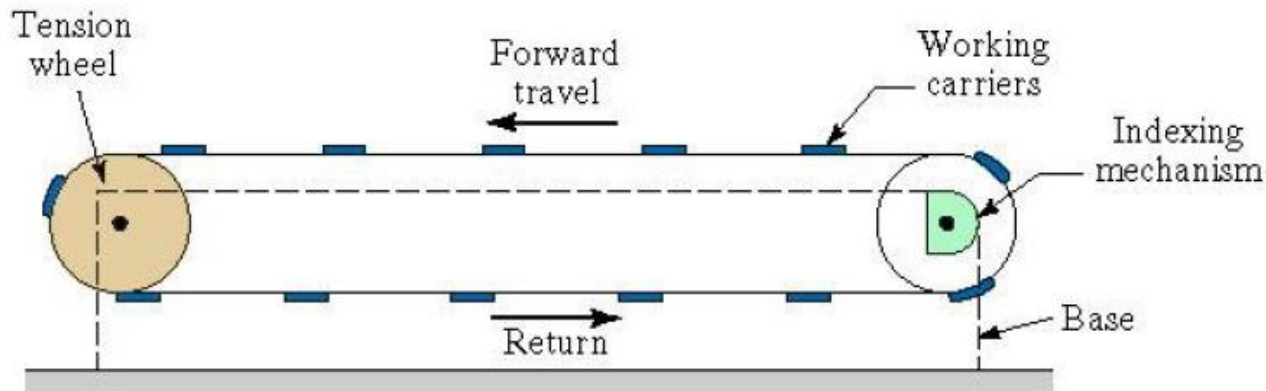


Figure 1.19 Chain drive conveyor

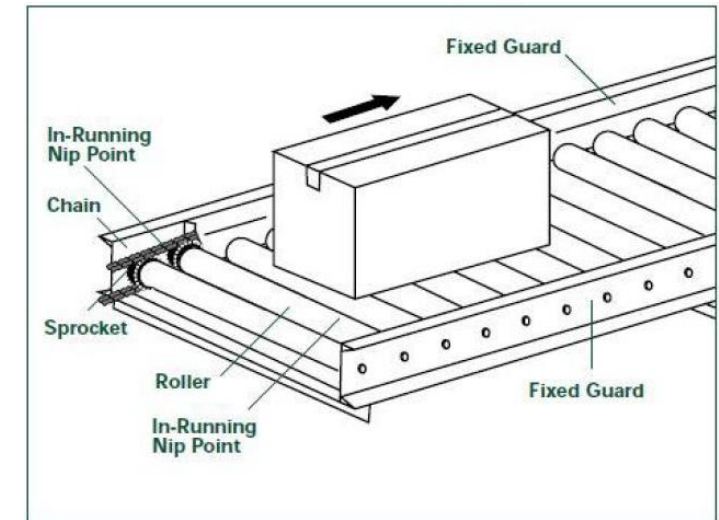


Figure 1.18 Powered roller conveyor

Rotary transfer mechanisms

a. Rack and pinion

- This mechanism is simple but is not suitable to the high-speed operation
- Piston to drive the rack, which causes the pinion gear and attached indexing table to rotate

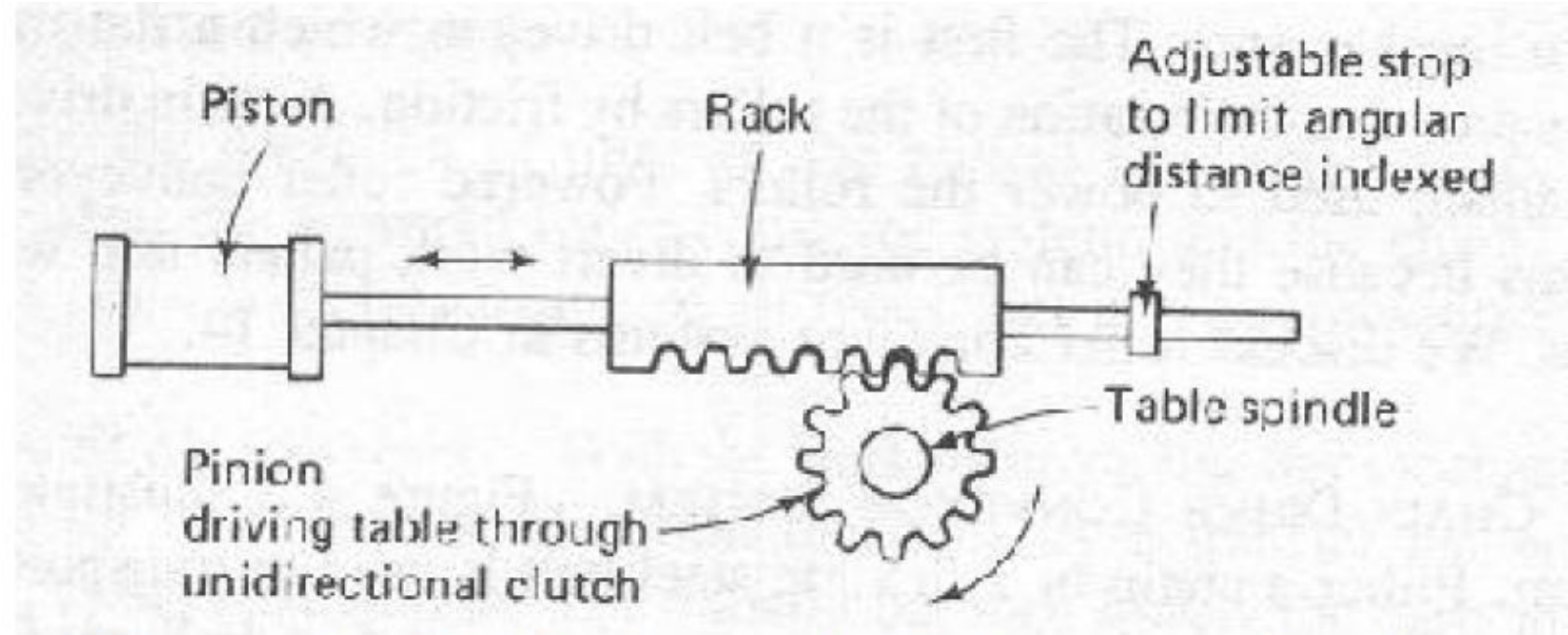
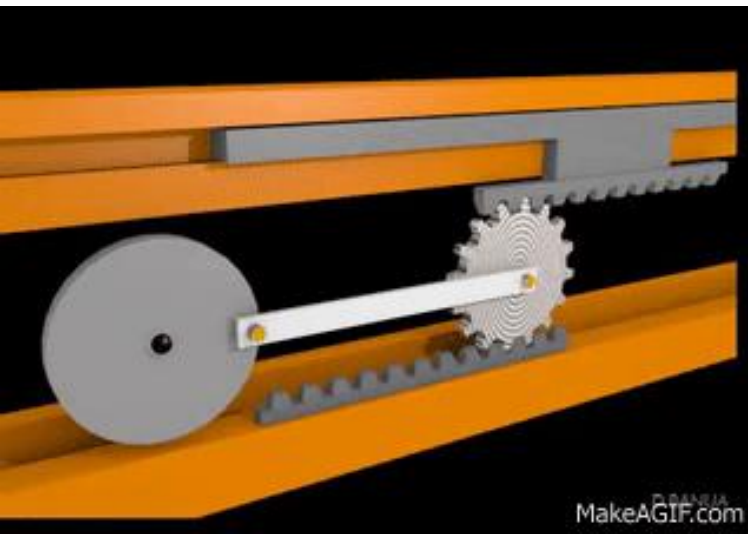
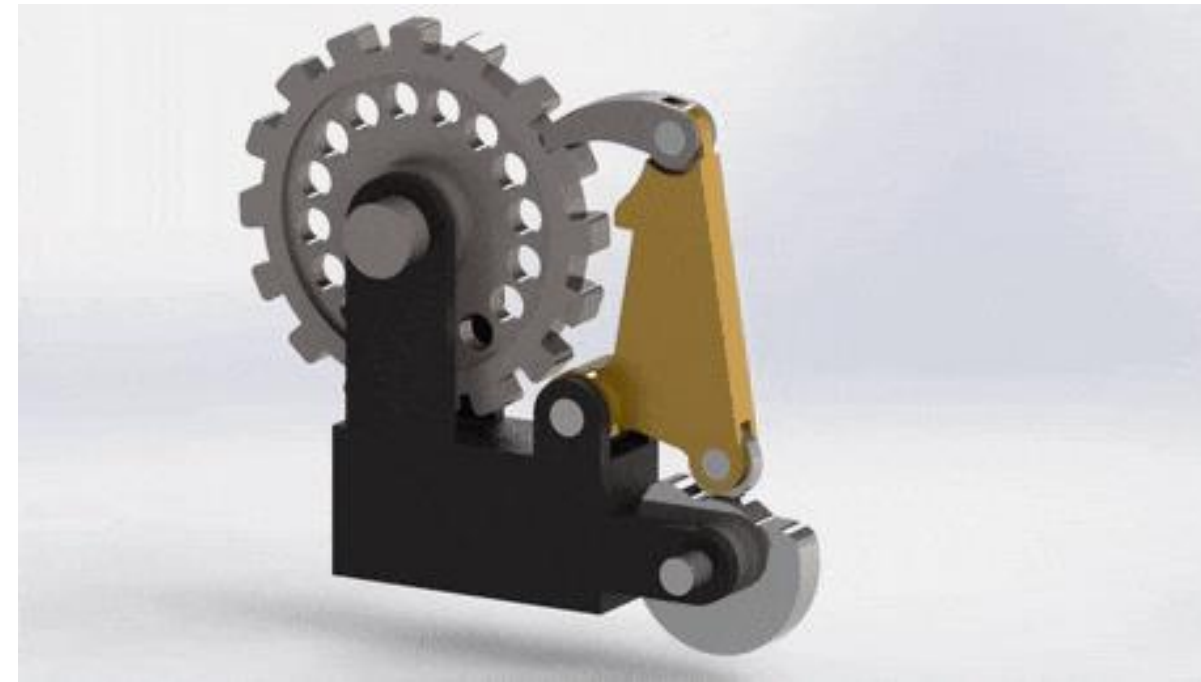
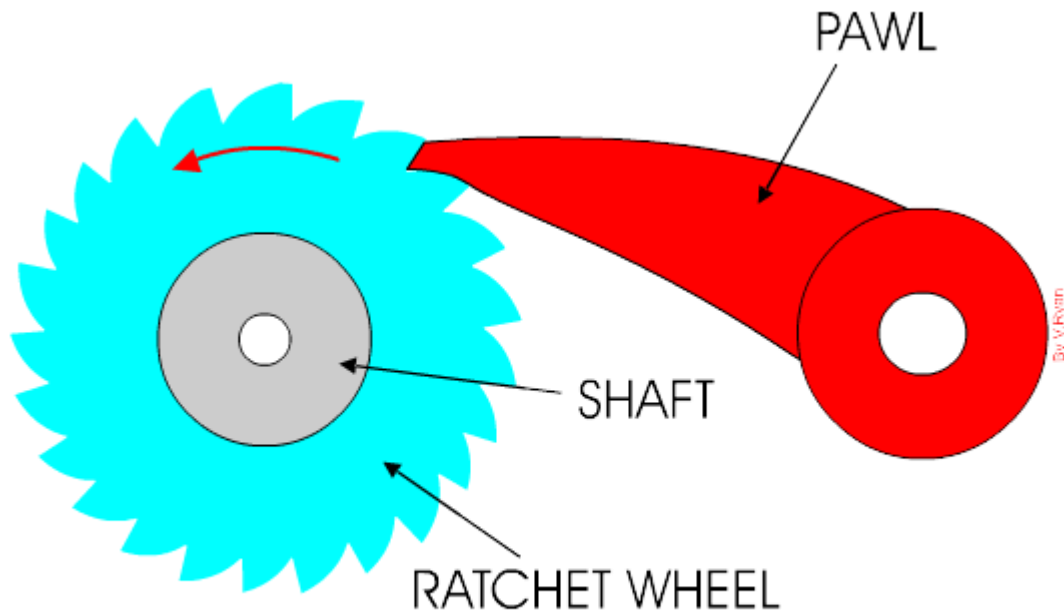
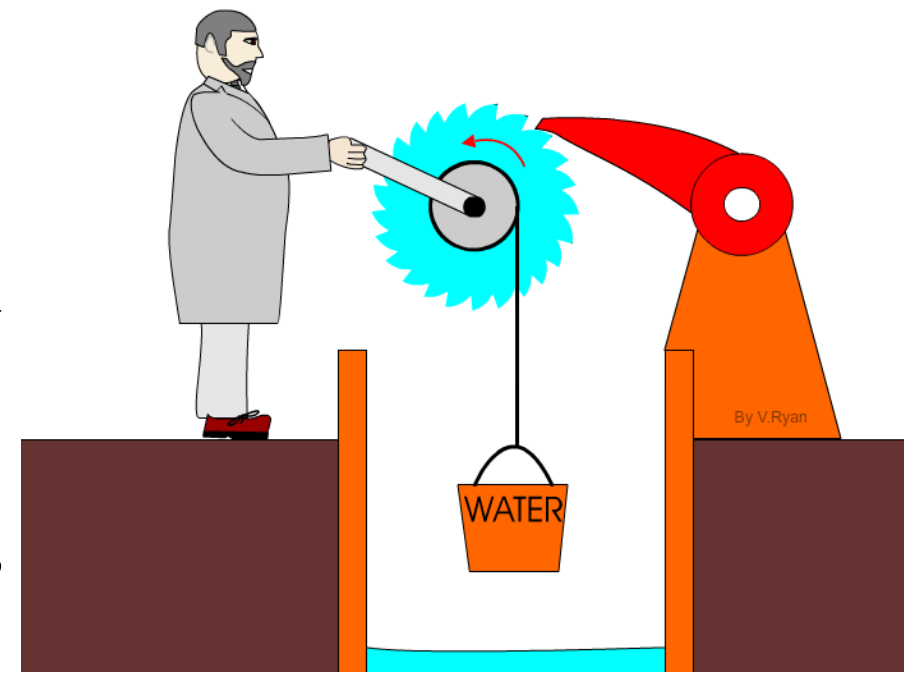


Figure 1.20 Rack and pinion mechanism

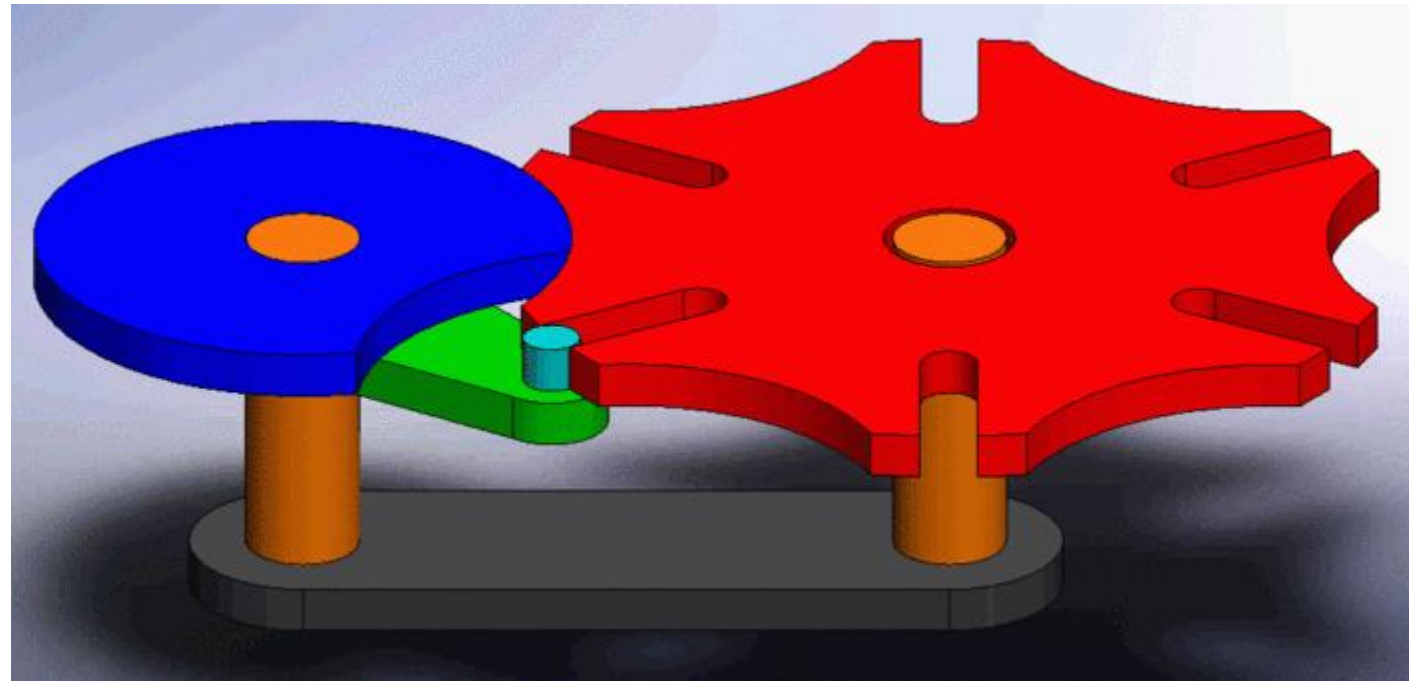
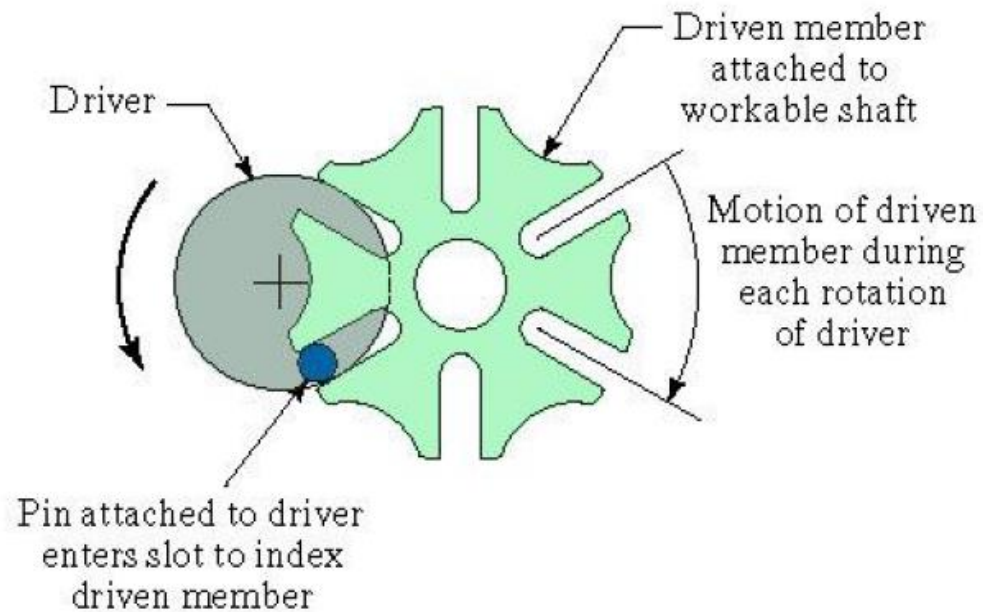
b. Ratchet and pawl

- A ratchet is a device that allows linear or rotary motion in only one direction, while preventing motion in the opposite direction.
- Ratchets consist of a gearwheel and a pivoting spring loaded finger called a pawl that engages the teeth



c. Geneva mechanism

- The two previous mechanisms convert a linear motion into a rotational motion.
- The Geneva mechanism uses a continuously rotating driver to index the table



d. CAM Mechanisms

- Various forms of cam mechanism provides accurate and reliable method of indexing the dial

